AERIS Expert Review Packet

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.065 / 0.0152157
Time	2026-07-21 22:00 UTC
Location	29.5831, -95.0156
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	6 / 646	29.01 to 100; mean 66.7901	41.9 at 2026-07-21 21:55Z; dt 5 min
asos	temperature	degC	6 / 646	23.2222 to 39; mean 30.732	36 at 2026-07-21 21:55Z; dt 5 min
asos	wind_direction	degree	6 / 740	0 to 360; mean 197.122	60 at 2026-07-21 21:55Z; dt 5 min
asos	wind_speed	m/s	6 / 778	0 to 8.2311; mean 2.6979	2.57222 at 2026-07-21 21:55Z; dt 5 min
noaa_gfs	gh_500	m	6 / 72	5915.17 to 5943.33; mean 5928.29	5927 at 2026-07-22 00:00Z; dt 120 min
noaa_gfs	pbl_height	m	6 / 72	17.7446 to 2017.14; mean 546.152	547.443 at 2026-07-22 00:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	6 / 72	29.3968 to 67.0225; mean 42.1369	41.7588 at 2026-07-22 00:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	6 / 72	1004.59 to 1013.76; mean 1009.67	1009.39 at 2026-07-22 00:00Z; dt 120 min
noaa_gfs	t_850	degC	6 / 72	20.702 to 24.3609; mean 22.2455	21.4217 at 2026-07-22 00:00Z; dt 120 min
noaa_gfs	u_10m	m/s	6 / 72	-4.5306 to 4.0994; mean 1.2261	0.797161 at 2026-07-22 00:00Z; dt 120 min
noaa_gfs	v_10m	m/s	6 / 72	-3.59026 to 4.8277; mean 0.3004	2.16704 at 2026-07-22 00:00Z; dt 120 min
openaq	ozone	ppm	15 / 973	-0.004 to 0.094; mean 0.027	0.065 at 2026-07-21 22:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 156	23 to 98; mean 45.7564	56 at 2026-07-21 22:00Z; dt 0 min
openaq	pm25	ug/m3	64 / 2408	0.3 to 29.4; mean 7.9339	4.7 at 2026-07-21 22:00Z; dt 0 min
openweather	cloud_cover	percent	3 / 216	0 to 100; mean 20.7269	4 at 2026-07-21 22:01Z; dt 1.9 min
openweather	humidity	percent	3 / 216	43 to 96; mean 70.3843	45 at 2026-07-21 22:01Z; dt 1.9 min
openweather	precipitation	mm/h	3 / 216	0 to 11.53; mean 0.1791	0 at 2026-07-21 22:01Z; dt 1.9 min
openweather	pressure	hPa	3 / 216	1007 to 1015; mean 1011.19	1010 at 2026-07-21 22:01Z; dt 1.9 min
openweather	temperature	degC	3 / 216	24.14 to 40.07; mean 31.4129	37.18 at 2026-07-21 22:01Z; dt 1.9 min
openweather	wind_direction	degree	3 / 216	0 to 359; mean 222.917	109 at 2026-07-21 22:01Z; dt 1.9 min
openweather	wind_speed	m/s	3 / 216	0.4 to 5.9; mean 2.2754	0.81 at 2026-07-21 22:01Z; dt 1.9 min
purpleair	pm25	ug/m3	59 / 4230	0 to 149.3; mean 10.0978	8.9 at 2026-07-21 22:00Z; dt 0 min
sentinel5p	s5p_aer_ai_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0262761 to 0.0312825; mean 0.028	0.0265506 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_co_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000212265 to 0.000396712; mean 0.0003	0.000224886 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_no2_column	mol/m^2	3 / 3	5.0358e-05 to 5.87309e-05; mean 0.0001	5.87309e-05 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_no2_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_o3_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-8.60324e-06 to 4.32022e-05; mean 0	3.4166e-05 at 2026-07-21 19:34Z; dt 145.4 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-21 19:34Z; dt 145.4 min
tceq	co	ppm	3 / 192	0 to 0.7; mean 0.1578	0.1 at 2026-07-21 22:00Z; dt 0 min
tceq	no2	ppb	11 / 658	-2.6 to 44.3; mean 6.2827	2.7 at 2026-07-21 22:00Z; dt 0 min
tceq	so2	ppb	3 / 186	-0.6 to 6.9; mean 0.0688	0.3 at 2026-07-21 22:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-20 06Z 94.7, 12Z 74.78, 18Z 49.43, 07-21 00Z 75.63, 06Z 92.45, 12Z 73.74, 18Z 39.69, 07-22 00Z 60.95, 06Z 79.14, 12Z 59.63, 18Z 42.15, 07-23 00Z 70.41, 06Z 75.68
asos	temperature	07-20 06Z 24.68, 12Z 29.15, 18Z 34.3, 07-21 00Z 29.31, 06Z 25.2, 12Z 29.02, 18Z 35.4, 07-22 00Z 30.81, 06Z 27.01, 12Z 31.64, 18Z 37.78, 07-23 00Z 31.34, 06Z 29.57
asos	wind_direction	07-20 06Z 138.6, 12Z 220.5, 18Z 157.7, 07-21 00Z 182.2, 06Z 189.4, 12Z 261.5, 18Z 218.6, 07-22 00Z 201.6, 06Z 237, 12Z 289.8, 18Z 212.8, 07-23 00Z 118.9, 06Z 52.79
asos	wind_speed	07-20 06Z 1.403, 12Z 2.446, 18Z 2.292, 07-21 00Z 3.201, 06Z 2.219, 12Z 3.508, 18Z 2.36, 07-22 00Z 2.476, 06Z 2.982, 12Z 3.928, 18Z 2.666, 07-23 00Z 1.972, 06Z 2.776
noaa_gfs	gh_500	07-20 12Z 5920, 18Z 5928, 07-21 00Z 5923, 06Z 5926, 12Z 5916, 18Z 5931, 07-22 00Z 5929, 06Z 5934, 12Z 5926, 18Z 5938, 07-23 00Z 5927, 06Z 5941
noaa_gfs	pbl_height	07-20 12Z 105.5, 18Z 1235, 07-21 00Z 505, 06Z 176.7, 12Z 184.9, 18Z 1188, 07-22 00Z 545.7, 06Z 109.9, 12Z 219.3, 18Z 1663, 07-23 00Z 314.4, 06Z 306.1
noaa_gfs	precipitable_water	07-20 12Z 30.33, 18Z 33.22, 07-21 00Z 37.3, 06Z 34.27, 12Z 31.33, 18Z 35.78, 07-22 00Z 40.72, 06Z 45.72, 12Z 52.55, 18Z 55.8, 07-23 00Z 61.66, 06Z 46.98
noaa_gfs	surface_pressure	07-20 12Z 1012, 18Z 1013, 07-21 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1011, 07-22 00Z 1008, 06Z 1009, 12Z 1009, 18Z 1009, 07-23 00Z 1006, 06Z 1008
noaa_gfs	t_850	07-20 12Z 22.62, 18Z 22.38, 07-21 00Z 21.28, 06Z 22.91, 12Z 22.46, 18Z 22.28, 07-22 00Z 21.85, 06Z 22.4, 12Z 21.68, 18Z 21.72, 07-23 00Z 23.75, 06Z 21.61
noaa_gfs	u_10m	07-20 12Z 1.645, 18Z 1.702, 07-21 00Z -1.465, 06Z 1.902, 12Z 2.739, 18Z 2.308, 07-22 00Z 0.8438, 06Z 2.517, 12Z 3.085, 18Z 0.662, 07-23 00Z -0.1673, 06Z -1.059
noaa_gfs	v_10m	07-20 12Z 0.5927, 18Z 0.6374, 07-21 00Z 3.017, 06Z 2.316, 12Z 0.2999, 18Z -0.7365, 07-22 00Z 0.5354, 06Z 1.391, 12Z -0.4914, 18Z -2.383, 07-23 00Z 1.481, 06Z -3.055

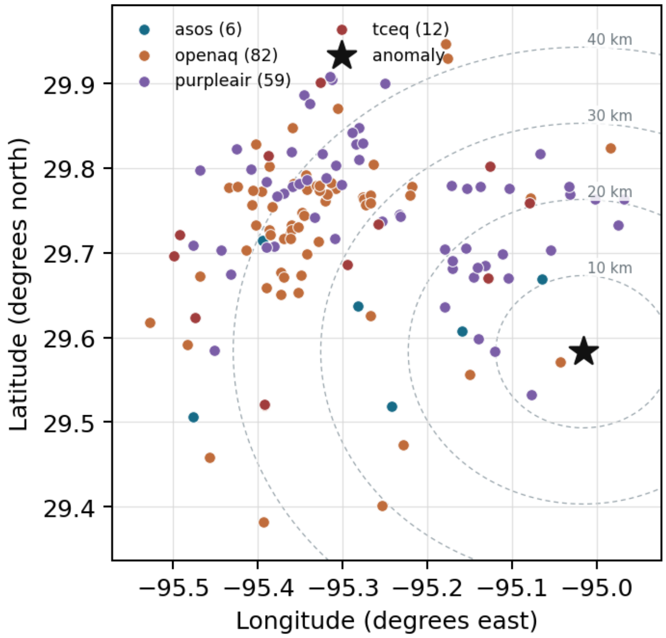
Source	Metric	Values
openaq	ozone	07-20 06Z 0.004667, 12Z 0.01069, 18Z 0.04479, 07-21 00Z 0.01837, 06Z 0.006118, 12Z 0.01152, 18Z 0.04518, 07-22 00Z 0.02902, 06Z 0.00619, 12Z 0.01574, 18Z 0.06606, 07-23 00Z 0.03856, 06Z 0.02694
openaq	pm10	07-20 06Z 45, 12Z 49.67, 18Z 50.75, 07-21 00Z 43.55, 06Z 39.83, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 31.13
openaq	pm25	07-20 06Z 8.987, 12Z 7.89, 18Z 6.619, 07-21 00Z 6.608, 06Z 7.377, 12Z 6.929, 18Z 5.735, 07-22 00Z 5.895, 06Z 6.012, 12Z 6.28, 18Z 9.156, 07-23 00Z 10.76, 06Z 18.17
openweather	cloud_cover	07-20 06Z 8.667, 12Z 3.278, 18Z 3.444, 07-21 00Z 3.706, 06Z 0.3158, 12Z 0.7059, 18Z 1.474, 07-22 00Z 4.389, 06Z 25.94, 12Z 13.06, 18Z 63.39, 07-23 00Z 80.88, 06Z 69.08
openweather	humidity	07-20 06Z 91.83, 12Z 79.56, 18Z 54.11, 07-21 00Z 74.35, 06Z 89.63, 12Z 80.12, 18Z 48.68, 07-22 00Z 62.89, 06Z 80.11, 12Z 70.67, 18Z 49.17, 07-23 00Z 74.12, 06Z 77
openweather	precipitation	07-20 06Z 0, 12Z 0, 18Z 0, 07-21 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.01, 07-23 00Z 2.265, 06Z 0
openweather	pressure	07-20 06Z 1014, 12Z 1015, 18Z 1012, 07-21 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1009
openweather	temperature	07-20 06Z 25.02, 12Z 29.07, 18Z 35.83, 07-21 00Z 30.67, 06Z 25.83, 12Z 28.88, 18Z 36.56, 07-22 00Z 31.93, 06Z 27.36, 12Z 31.33, 18Z 39.05, 07-23 00Z 31.87, 06Z 30.26
openweather	wind_direction	07-20 06Z 200.3, 12Z 261.9, 18Z 225.2, 07-21 00Z 193.6, 06Z 186, 12Z 268.3, 18Z 255.6, 07-22 00Z 224.1, 06Z 258.9, 12Z 283.7, 18Z 172.4, 07-23 00Z 221, 06Z 98.23
openweather	wind_speed	07-20 06Z 1.593, 12Z 1.838, 18Z 2.032, 07-21 00Z 2.891, 06Z 2.414, 12Z 2.092, 18Z 1.464, 07-22 00Z 2.726, 06Z 2.276, 12Z 2.136, 18Z 2.466, 07-23 00Z 2.711, 06Z 2.687
purpleair	pm25	07-20 06Z 12.09, 12Z 9.759, 18Z 7.661, 07-21 00Z 8.626, 06Z 9.612, 12Z 8.39, 18Z 7.889, 07-22 00Z 9.967, 06Z 6.817, 12Z 7.95, 18Z 11.08, 07-23 00Z 14.7, 06Z 21.95
sentinel5p	s5p_aer_ai_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	s5p_ch4_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	s5p_co_column	07-20 18Z 0.02628, 07-21 18Z 0.02656, 07-22 18Z 0.03123
sentinel5p	s5p_co_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	s5p_hcho_column	07-20 18Z 0.000213, 07-21 18Z 0.0002243, 07-22 18Z 0.0003967
sentinel5p	s5p_hcho_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	s5p_no2_column	07-20 18Z 5.835e-05, 07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	s5p_no2_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	s5p_o3_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	s5p_so2_column	07-20 18Z 1.73e-05, 07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06
sentinel5p	s5p_so2_granule_available	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
tceq	co	07-20 06Z 0.3667, 12Z 0.2125, 18Z 0.1722, 07-21 00Z 0.2222, 06Z 0.1222, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1267
tceq	no2	07-20 06Z 6.273, 12Z 6.184, 18Z 6.378, 07-21 00Z 6.763, 06Z 5.348, 12Z 5.514, 18Z 4.918, 07-22 00Z 10.8, 06Z 6.498, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.194, 06Z 4.602
tceq	so2	07-20 06Z -0.3333, 12Z -0.175, 18Z 1.906, 07-21 00Z 1.213, 06Z -0.2778, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3

Per-site averages

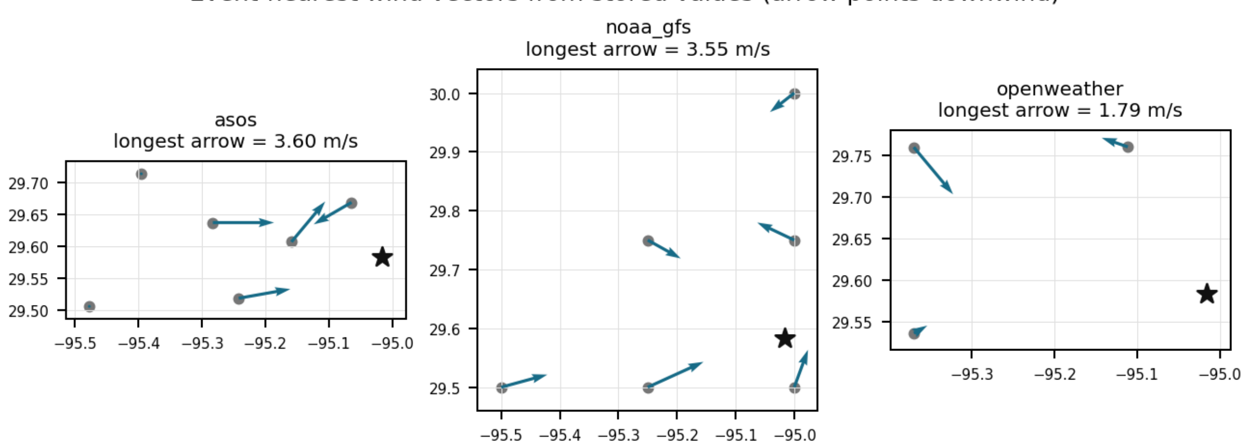
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.66 km) 68.71, EFD (14.095 km) 64.32, LVJ (23.006 km) 70.78, HOU (26.492 km) 66.05, MCJ (39.448 km) 61.72, AXH (45.438 km) 69.25
asos	temperature	T41 (10.66 km) 30.64, EFD (14.095 km) 31.44, LVJ (23.006 km) 30.06, HOU (26.492 km) 31, MCJ (39.448 km) 31.48, AXH (45.438 km) 29.99
asos	wind_direction	T41 (10.66 km) 198.7, EFD (14.095 km) 206.6, LVJ (23.006 km) 189.7, HOU (26.492 km) 211.9, MCJ (39.448 km) 217.7, AXH (45.438 km) 168.7
asos	wind_speed	T41 (10.66 km) 3.004, EFD (14.095 km) 3.36, LVJ (23.006 km) 1.818, HOU (26.492 km) 3.344, MCJ (39.448 km) 3.605, AXH (45.438 km) 1.424
noaa_gfs	gh_500	gfs:29.50,-95.00 (9.363 km) 5927, gfs:29.75,-95.00 (18.62 km) 5928, gfs:29.50,-95.25 (24.486 km) 5927, gfs:29.75,-95.25 (29.28 km) 5929, gfs:30.00,-95.00 (46.382 km) 5930, gfs:29.50,-95.50 (47.763 km) 5928
noaa_gfs	pbl_height	gfs:29.50,-95.00 (9.363 km) 500.7, gfs:29.75,-95.00 (18.62 km) 552.8, gfs:29.50,-95.25 (24.486 km) 496.1, gfs:29.75,-95.25 (29.28 km) 861.4, gfs:30.00,-95.00 (46.382 km) 382, gfs:29.50,-95.50 (47.763 km) 483.9
noaa_gfs	precipitable_water	gfs:29.50,-95.00 (9.363 km) 42.73, gfs:29.75,-95.00 (18.62 km) 42.56, gfs:29.50,-95.25 (24.486 km) 42.33, gfs:29.75,-95.25 (29.28 km) 41.66, gfs:30.00,-95.00 (46.382 km) 42.06, gfs:29.50,-95.50 (47.763 km) 41.47
noaa_gfs	surface_pressure	gfs:29.50,-95.00 (9.363 km) 1011, gfs:29.75,-95.00 (18.62 km) 1010, gfs:29.50,-95.25 (24.486 km) 1010, gfs:29.75,-95.25 (29.28 km) 1009, gfs:30.00,-95.00 (46.382 km) 1009, gfs:29.50,-95.50 (47.763 km) 1009
noaa_gfs	t_850	gfs:29.50,-95.00 (9.363 km) 22.19, gfs:29.75,-95.00 (18.62 km) 22.2, gfs:29.50,-95.25 (24.486 km) 22.18, gfs:29.75,-95.25 (29.28 km) 22.27, gfs:30.00,-95.00 (46.382 km) 22.33, gfs:29.50,-95.50 (47.763 km) 22.3
noaa_gfs	u_10m	gfs:29.50,-95.00 (9.363 km) 1.575, gfs:29.75,-95.00 (18.62 km) 0.5752, gfs:29.50,-95.25 (24.486 km) 1.277, gfs:29.75,-95.25 (29.28 km) 1.118, gfs:30.00,-95.00 (46.382 km) 0.7827, gfs:29.50,-95.50 (47.763 km) 2.029
noaa_gfs	v_10m	gfs:29.50,-95.00 (9.363 km) 0.942, gfs:29.75,-95.00 (18.62 km) 0.2529, gfs:29.50,-95.25 (24.486 km) 0.6037, gfs:29.75,-95.25 (29.28 km) 0.02119, gfs:30.00,-95.00 (46.382 km) -0.458, gfs:29.50,-95.50 (47.763 km) 0.4404
openaq	ozone	5077791 (0.0 km) 0.02461, 332 (14.578 km) 0.03198, 2800 (21.045 km) 0.02518, 320 (24.779 km) 0.02771, 3214 (26.61 km) 0.02965, 339 (26.887 km) 0.03152, 2039 (28.555 km) 0.02672, 3378 (28.767 km) 0.02673, +7 more
openaq	pm10	271 (14.578 km) 53.79, 13380082 (28.767 km) 51.29, 15852419 (35.779 km) 36.63
openaq	pm25	14404954 (2.966 km) 7.462, 13955815 (13.303 km) 7.006, 268 (14.578 km) 17.96, 11638043 (23.947 km) 5.384, 14157975 (24.782 km) 8.91, 3379 (28.767 km) 14.09, 15539682 (28.768 km) 9.843, 8967123 (28.822 km) 11.02, +56 more
openweather	cloud_cover	grid:east (21.767 km) 18.11, grid:south (34.66 km) 23.49, grid:center (39.488 km) 20.58
openweather	humidity	grid:east (21.767 km) 74.11, grid:south (34.66 km) 70.33, grid:center (39.488 km) 66.71
openweather	precipitation	grid:east (21.767 km) 0.2261, grid:south (34.66 km) 0.2347, grid:center (39.488 km) 0.07653
openweather	pressure	grid:east (21.767 km) 1011, grid:south (34.66 km) 1011, grid:center (39.488 km) 1011
openweather	temperature	grid:east (21.767 km) 30.96, grid:south (34.66 km) 31.06, grid:center (39.488 km) 32.22
openweather	wind_direction	grid:east (21.767 km) 216.2, grid:south (34.66 km) 216.5, grid:center (39.488 km) 236
openweather	wind_speed	grid:east (21.767 km) 2.528, grid:south (34.66 km) 2.245, grid:center (39.488 km) 2.053
purpleair	pm25	2386 (8.161 km) 0.3597, 3298 (10.104 km) 10.79, 247283 (12.088 km) 11.17, 268165 (12.918 km) 11.97, 222815 (13.885 km) 10.51, 222593 (15.857 km) 10.58, 128007 (15.88 km) 5.008, 161689 (15.935 km) 10.82, +51 more
tceq	co	482011039 (14.558 km) 0.1063, 482011035 (28.764 km) 0.171, 482011052 (44.204 km) 0.1955
tceq	no2	482011050 (0.008 km) 2.986, 482011039 (14.558 km) 7.165, 482011015 (20.5 km) 9.998, 482010026 (26.626 km) 7.192, 482011035 (28.764 km) 7.61, 482010416 (29.317 km) 6, 480391004 (37.118 km) 3.982, 482011052 (44.204 km) 5.483, +3 more
tceq	so2	482011039 (14.558 km) 0.4781, 482011035 (28.764 km) -0.2917, 482010051 (44.584 km) -0.004839

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO_2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

Local-to-regional precursor accumulation under weak dispersion is supported because surface winds near the anomaly were light at about 0.8 to 2.6 m/s, modeled afternoon boundary-layer height was moderate rather than strongly ventilating at about 547 m near 00Z, and ozone was broadly elevated across the 18Z 6-hour period rather than appearing as a single isolated sensor spike.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

An ozone concentration anomaly of 0.065 ppm was measured at latitude 29.583, longitude -95.016 on July 21, 2026, at 22:00:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

Observational data from asos and openweather indicated strong solar heating with surface temperatures exceeding 36 degrees Celsius, clear skies, and light surface winds during the afternoon of July 21, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

Ambient nitrogen dioxide concentrations measured by TCEQ alongside satellite tropospheric formaldehyde and nitrogen dioxide column abundances detected by Sentinel-5P confirm the local availability of precursor pollutants required for photochemical ozone generation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO_2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

A Houston Ship Channel industrial precursor contribution is plausible but only partially supported because the anomaly site lies in eastern Houston where nearby winds briefly shifted to easterly or east-southeasterly directions around the event, Sentinel-5P showed relatively elevated same-day NO_2 and SO_2 columns, but nearby surface NO_2, SO_2, and CO at 22:00Z were low enough to argue against a strong fresh combustion plume directly over the monitor at the anomaly hour.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Light surface wind speeds averaging around 2.6 m/s recorded by ASOS and moderate planetary boundary layer heights modeled by GFS reduced local ventilation, allowing ozone precursors to accumulate.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

The PM2.5 levels were higher than usual due to increased particulate matter emissions from nearby industrial activities or wildfires.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

The anomaly occurred on July 21st at around 22:01:54+00:00 (1.9 min away) and is located approximately 21.767 km east of the grid center.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

A strong wind pattern was observed on July 21st, potentially dispersing pollutants in the air and affecting air quality readings.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

Weak dispersion likely amplified ozone concentrations near eastern Houston because local wind speeds were light and boundary-layer conditions were consistent with limited ventilation compared with a stronger transport regime.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Photochemical ozone production is strongly supported because the ozone anomaly occurred during very hot, dry, mostly cloud-free late-afternoon conditions, with ozone reaching 0.065 ppm at 22:00Z while nearby temperature was about 36 to 37.2 C, relative humidity was about 42 to 45 percent, cloud cover was about 4 percent, and no precipitation was reported near the event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

The PM2.5 levels are elevated at 8.9 ug/m3, which is higher than the mean of 10.0978 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

High surface temperatures reaching 36.0°C measured by ASOS and minimal cloud cover under 5 percent reported by OpenWeather around 22:00 UTC created conditions conducive to rapid photochemical ozone formation measured by OpenAQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

Meteorological observations near the anomaly time showed approximately 36.0 to 37.18 degC temperature, about 41.9 to 45.0 percent humidity, 4.0 percent cloud cover, and light winds of about 0.81 to 2.57 m/s near the anomaly location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

A surface temperature of 36.0 degrees Celsius and a relative humidity of 41.9 percent were observed near the anomaly site at 21:55:00 UTC on July 21, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The ozone concentration at the anomaly was 0.065 ppm, compared with an expected value of 0.015215686274509806 ppm, corresponding to a z-score of 5.742374883976655 and labeled severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

Ground monitoring networks operated by tceq detected ambient nitrogen dioxide concentrations that supplied the required chemical precursors for daytime photolysis and ozone production.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

An openaq ground monitor recorded a severe surface ozone concentration of 0.065 ppm in Houston at 22:00 UTC on July 21, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

A localized urban-industrial precursor contribution near the Houston Ship Channel is plausible but only partially supported because satellite NO2 and SO2 showed some enhancement while surface NO2, CO, and SO2 near the anomaly were not strongly elevated at the ozone peak.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

Emissions from the eastern Houston industrial and ship-channel corridor are a plausible contributing precursor source because the anomaly was located in eastern Houston and satellite sulfur dioxide and nitrogen dioxide columns were elevated earlier that day even though nearby surface NO2 was low at the ozone peak.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

The air quality anomaly was characterized by a significant increase in PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

The pollutant anomaly was ozone measured by openaq at 2026-07-21T22:00:00+00:00 near 29.583, -95.016 in the Houston area.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

The ozone level of 0.065 ppm on July 21, 2026, at 22:00:00 UTC represented a z-score elevation of 5.74 above the expected mean value of 0.015 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

The temperature was unusually high on July 21st, causing increased emissions from industrial and vehicular sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

Strong photochemical ozone formation likely contributed to the Houston ozone spike because the anomaly occurred in very hot, dry, mostly cloud-free late-afternoon conditions that favor rapid ozone production from NOx and VOC precursors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Weak dispersion and mesoscale recirculation likely contributed to the Houston ozone spike because surface winds near the anomaly were light and wind direction varied between nearby observations, which can trap and re-transport ozone precursors and aged air over the urban-industrial area.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

The temperature and wind speed were within normal ranges during the time of the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

Emissions of nitrogen oxides and reactive organic precursor gases from the Houston industrial corridor supplied the necessary chemical reactants for local photochemical ozone generation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

Weak surface winds and atmospheric stagnation reduced precursor ventilation, allowing ozone and its precursors to accumulate locally during peak afternoon hours.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

There was no significant increase in CO, NO2, or SO2 levels that could have contributed to the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The ozone anomaly near eastern Houston at 2026-07-21 22:00 UTC was most likely driven primarily by enhanced afternoon photochemical ozone production under hot, relatively dry, mostly clear conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

High ambient temperatures exceeding 36 °C combined with clear skies enhanced photolysis rates and accelerated daytime photochemical ozone synthesis from precursor gases in Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.